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Using mathematical morphology to discover repeated patterns in music

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This article presents an application of mathematical morphology to the discovery of repeated patterns in a multidimensional representation of music. In particular, we demonstrate the possibility of discovering patterns from their *onsets*, defined as the translations of the pattern to obtain its occurrences in musical data. We first prove this result for *fully compact* patterns, meaning that they contain all dataset points in their convex hull. We then prove that patterns can be discovered from their onsets if and only if they are *maximal with respect to their translational equivalence classes* (*MTEC*), meaning that no points can be added to it without decreasing the number of occurrences. Moreover, we prove that applying twice the morphological erosion completes any pattern by adding the points required to make it MTEC, and yields an MTEC pattern and their onsets linked by erosion. Finally, we apply these results to musical examples, illustrating how morphological operators can be used as analytical tools for describing musical data.

Keywords: pattern discovery; mathematical morphology; erosion; geometric pattern discovery in music; music analysis; point-set algorithms; discovering a musical pattern from its onsets; conjugate patterns; SIA; MTP; MTEC

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1. Introduction

In this article, we propose several mathematical results based on the theory of mathematical morphology for discovering repeated patterns in musical data. In particular, these results provide important properties about repeating patterns in a discrete set of points and they offer original properties of the basic operators of mathematical morphology, such as erosion. Since mathematical morphology is generally applied to image processing, we might then ask: why is this theory particularly adapted for the discovery of musical patterns? This is partly due to the fact that morphological operations involve translating a *structuring element* to produce a result that depends on how it intersects with musical data. Therefore, by assigning a musical meaning to the structuring element, it is possible to extract certain characteristics from musical data. This was first achieved with a pattern-structuring element to extract occurrences of a given pattern (Karvonen and Lemström 2008; Karvonen et al. 2010), or to extract harmonic components (Lascabettes, Bloch, and Agon 2020), for example with a major-structuring element that extracts major chords. Here, we propose an onset-structuring element[1] to discover patterns, as we reveal that patterns and their onsets are linked by morphological

[1]We define the *onsets* of a pattern as the set of points that correspond to the translations of the pattern to obtain its occurrences in musical data. The onsets of a pattern can be interpreted as the beginnings of the occurrences of a pattern in musical data.

operators. We then formalize the problem of *discovering a musical pattern from its onsets* and provide a sufficient condition for this discovery when the patterns cover the data and are *fully compact*, a local topological property, as well as a necessary and sufficient condition when the patterns are *maximal with respect to their translational equivalence classes* (MTEC), a global topological property which means that no points can be added to the pattern without decreasing its number of occurrences. We then prove that applying morphological erosion twice adds the necessary points to a pattern to obtain the smallest MTEC that contains it. Finally, we produce in this paper original mathematical results that start from musical motivations and can be applied to other types of discrete point set data.

The remainder of this article is structured as follows. We introduce in Section 2 the notions of mathematical morphology that are useful in this article. In Section 3, we show that the results presented in this article are a continuation of previous work on the musical pattern discovery task. We present in Section 4 the problem of discovering patterns from their onsets. In Section 4.2, we prove that this problem can be solved under the assumption that the patterns cover the data and are fully compact, and in Section 4.3, we demonstrate that the patterns discovered by their onsets correspond exactly to the MTEC patterns. We then explore the pairs of patterns and their onsets in Section 5, and reveal that it is always possible to obtain such a pair from a given pattern, as the morphological erosion adds any points to a pattern required to make it MTEC. We then apply the results developed in this article to musical examples in Section 6, and illustrate how morphological operators can be used as analysis tools to describe musical data. Finally, we summarize the main results and conclude this article in Section 7.

2. Notations of binary mathematical morphology

Mathematical morphology, in its deterministic part, is an algebraic theory of shape invented in the 1960s by Matheron (1967, 1975) and Serra (1982, 1988), and widely used in image analysis. It changes the shapes contained in the images with non-linear and non-reversible transformations. We present in this section the concepts of mathematical morphology that are useful in this article, in order for it to be self-contained. In particular, only the fundamental notions of binary mathematical morphology are introduced here and we only require the case where the principal operators of binary mathematical morphology are applied in $\mathbb{R}^n$. The majority of morphological operators, in the deterministic setting, result from two basic ones: the *dilation* from the *Minkowski addition* $\oplus$ (Minkowski 1903) and the *erosion* from the *subtraction* $\ominus$ (Hadwiger 1950). It is then possible to generate other morphological operators using compositions, unions, intersections, or complements.

Definition 2.1 (Dilation and erosion) Let $X, S \subseteq \mathbb{R}^n$, the *dilation* $\delta_S(X)$ and *erosion* $\varepsilon_S(X)$ of X by S are defined by the following.

$$\delta_S(X) = X \oplus S = \{x + s \mid x \in X, s \in S\} \tag{1}$$

$$\varepsilon_S(X) = X \ominus S = \{x \in \mathbb{R}^n \mid S_x \subseteq X\} \tag{2}$$

where S_x is the translate of $S \subseteq \mathbb{R}^n$ by $x \in \mathbb{R}^n$ defined by $S_x = \{s + x \mid s \in S\}$. In this case, S is called a *structuring element*. By composing these two operations, we obtain the two other fundamental operators of mathematical morphology: *opening* and *closing*.

Definition 2.2 (Opening and closing) Let $X, S \subseteq \mathbb{R}^n$, the *opening* $\gamma_S(X)$ and the *closing* $\varphi_S(X)$ of X by S are defined by the following.

$$\gamma_S(X) = X \circ S = \delta_S(\varepsilon_S(X)) = \bigcup \{S_x \mid x \in \mathbb{R}^n \wedge S_x \subseteq X\} \tag{3}$$

$$\varphi_S(X) = X \bullet S = \varepsilon_S(\delta_S(X)) \tag{4}$$

These four operations satisfy many different mathematical properties, and we present the essential ones that are required in this article, assuming $X, S, S' \subseteq \mathbb{R}^n$.

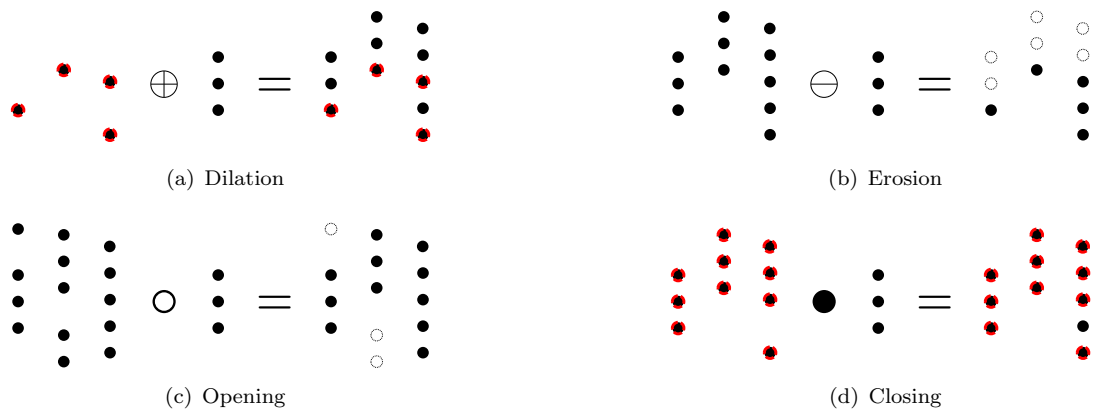
- They are all *increasing* (i.e. $X \subseteq X' \Rightarrow f_S(X) \subseteq f_S(X')$, where f_S denotes one of the four operations).
- The dilation is *increasing according to the structuring element* (i.e. $S \subseteq S' \Rightarrow \forall X,\ \delta_S(X) \subseteq \delta_{S'}(X)$) while the erosion is *decreasing according to the structuring element* (i.e. $S \subseteq S' \Rightarrow \forall X,\ \varepsilon_{S'}(X) \subseteq \varepsilon_S(X)$).
- The opening and closing are both *idempotent* (i.e. $\gamma_S \circ \gamma_S = \gamma_S$ and $\varphi_S \circ \varphi_S = \varphi_S$). This means that they only need to be applied once and there is no need to repeat them. Consequently, they are *morphological filters*, defined as increasing and idempotent operators.
- If and only if the origin belongs to S, the dilation is *extensive* (i.e. $O_{\mathbb{R}^n} \in S \iff X \subseteq \delta_S(X)$) and the erosion is *anti-extensive* (i.e. $O_{\mathbb{R}^n} \in S \iff \varepsilon_S(X) \subseteq X$). However, in all cases, the opening is *anti-extensive* (i.e. $\gamma_S(X) \subseteq X$) and the closing is *extensive* (i.e. $X \subseteq \varphi_S(X)$).
- The dilation is *commutative* ($\delta_S(S') = \delta_{S'}(S)$).

Finally, if we consider that the origin of space belongs to S and that X is a piece of music, among the four basic operations of mathematical morphology, erosion and opening are *analysis operators* (extracting musical data), while dilation and closing are *generation operators* (enriching the music). Since we focus in this paper on the discovery of repeated patterns in music, i.e. on music analysis and not generation, we therefore mainly use the opening and the erosion, and demonstrate that these two operations are particularly adapted to this problem.

Figure 1 illustrates the application of these four basic operations of mathematical morphology to a discrete set of points, which can be interpreted as points in $\mathbb{R}^2$. Therefore, these four operations can be interpreted as follows:

- **Dilation:** Union of all translations of S to positions in X.
- **Erosion:** All positions at which S occurs in X.
- **Opening:** Union of all occurrences of S in X.
- **Closing:** All positions at which S occurs in the union of all translations of S to positions in X.

We have chosen to illustrate these operations on a discrete set of points and not on continuous shapes, as is usually done, since this is how we apply these operations in this article. Indeed, while mathematical morphology has been widely applied to image processing, analysis and understanding, there are only few direct applications of this theory, in particular in its algebraic setting, to symbolic representations of music. This is partly due to the different nature of the patterns. On the one hand, the objects associated with digital image processing are defined on a discrete grid, endowed with a discrete connectivity, and are often defined as connected sets. On the other hand, those associated



(a) Dilation (b) Erosion

(c) Opening (d) Closing

Figure 1. Application of the basic operators of mathematical morphology to a set of points. The origin of the structuring element is located at its lowest point. For dilation (a) and closing (d) operations, the points of the starting set are framed in red dotted lines, to distinguish them from the points added after these operations. For erosion (b) and opening (c) operations, the points that are removed after these operations are indicated by dotted black lines with transparent interiors on the resulting set.

with symbolic music are sparse, in the sense that they are often not connected, according to the underlying connectivity of the space of representation. It is therefore necessary to adapt the morphological tools and the expected results when applying this theory to symbolic representations of music. This is what we do in this article, by proposing new definitions, such as the criterion that a pattern is fully compact (see Section 4.2), which can be considered as the counterpart of the connectivity used in image processing, or by assigning a musical meaning to morphological operations and results, such as distinguishing the role of onsets and musical patterns with the morphological erosion. This formalization allows the operations of mathematical morphology to be applied to symbolic representations of music, producing mathematical results that are useful for discovering musical patterns.

3. Context of the proposed method

The task of discovering musical patterns in a symbolic representation of music consists in automatically identifying repeated or almost-repeated patterns. This provides a more compact description of the data, due to the repetition present in music. Although many musical pattern discovery algorithms have been developed in the past few years, it remains a difficult task. There are two main approaches to discovering patterns in a symbolic representation of music: *string-based approaches* and *geometric approaches* (Janssen et al. 2014). The first attempts to discover patterns in music were based on a string representation of music. In this case, musical data is represented by a sequence of symbols. From this representation, the goal is to find subsequences that repeat identically or nearly identically (Hsu, Chen, and Liu 1998; Smith and Medina 2001). The string representation is readily adapted for monophonic music but might face difficulties when handling polyphonic music, especially when the representation does not contain (reliable) information about how the music is structured into voices or parts. This was one of the main motivations behind Meredith, Lemström, and Wiggins (2002) development of a geometric approach to pattern discovery in music, based on multidimensional point-set representations rather than strings. The originality of this approach is to represent musical data in an n-dimensional space, usually $\mathbb{R}^n$, where each point corresponds to a musical note and

each component of the space corresponds to a characteristic of the note. For example, in an n-dimensional space, the first component can represent the note onset, the second component the pitch of the note, the third component its duration, and other components associated with other parameters can be added according to the data. Based on this representation, several algorithms, called *point-set algorithms*, have been developed. They can be classified into four categories:

- Algorithms that discover repeated patterns in a dataset: SIA discovers all *maximal translatable patterns* (MTPs) (Meredith, Wiggins, and Lemström 2001; Meredith, Lemström, and Wiggins 2002), SIATEC the occurrences of these MTPs (Meredith, Wiggins, and Lemström 2001; Meredith, Lemström, and Wiggins 2002; Meredith 2006), SIAR the MTPs given a sliding windows (Collins 2011), and SIACT the sub-patterns of MTPs that are compact (Collins et al. 2010).
- Algorithms for pattern matching: SIAM discovers the maximal translatable matches of a given pattern in a dataset (Meredith, Wiggins, and Lemström 2001; Wiggins, Lemström, and Meredith 2002; Meredith, Lemström, and Wiggins 2002).
- Algorithms that discover the most interesting patterns to describe the dataset: COSIATEC covers the dataset without overlaps (Meredith, Lemström, and Wiggins 2002; Meredith 2006), SIATECCompress covers the dataset with overlaps (Meredith 2013, 2015), and Forth's algorithm does not cover the entire dataset and allows for overlaps (Forth and Wiggins 2009; Forth 2012).
- Algorithms that discover patterns with variations: More recently, Meredith (2023, 2024) has also proposed an algorithm for discovering all the *maximal transformable patterns* in a dataset with respect to a user-defined class of transformations such as augmentation, diminution, inversion and retrograde. Also, Laaksonen and Lemström (2021), Laaksonen, Lemström, and Björklund (2022) and Laaksonen and Lemström (2024) have developed methods for finding pattern repetitions under time-warp invariances in musical data.

The results we propose in this article may be useful for the third category of algorithm, as we are interested in the possibility of covering the dataset with patterns that repeat at regular time intervals. The novelty of our approach is the use of mathematical morphology, and we reveal that it provides original mathematical results for the pattern discovery task in a multidimensional representation of music. Therefore, in this paper, musical data are represented by a subset X of $\mathbb{R}^n$, and the objective is to discover a family $\mathcal{F} = (P_i)_{i \in I}$ of musical patterns, such that I is an index set, none of the patterns P_i are singletons, and the union of the openings of X by these patterns is approximately equal to X:

$$\gamma_{\mathcal{F}}(X) \approx X \tag{5}$$

where the relation $\approx$ can be interpreted by a relatively high proportion of common points between $\gamma_{\mathcal{F}}(X)$ and X, and the opening by $\mathcal{F}$ is defined by the union of the openings by the elements of $\mathcal{F}$:

$$\gamma_{\mathcal{F}}(X) = \bigcup_{P_i \in \mathcal{F}} \gamma_{P_i}(X) \tag{6}$$

In this case, $\gamma_{\mathcal{F}}$ is an opening in a broader algebraic sense, because it is an increasing, idempotent and anti-extensive operation. But how do we obtain the elements of $\mathcal{F}$? It has already been proved that there are links between point-set algorithms and mathematical

morphology. For example, MTPs, which are the basis of point-set algorithms and defined by $MTP(v, X) = \{x \in X \mid x + v \in X\}$ with $v \in \mathbb{R}^n$ and $X \subseteq \mathbb{R}^n$, can be expressed in a very simple way using the morphological erosion (Lascabettes 2023).

Lemma 3.1 (Characterization of MTP patterns with the erosion) *Let $X \subseteq \mathbb{R}^n$ and $v \in \mathbb{R}^n$. By denoting $\{0, v\}$ the bipoint composed of the origin and any point v, we have:*

$$MTP(v, X) = \varepsilon_{\{0,v\}}(X) \tag{7}$$

We therefore propose to exploit the property of morphological erosion as a generalization of MTPs. We demonstrate that, by selecting the appropriate structuring elements of the erosion, this operation discovers salient musical patterns. But how do we select the appropriate structuring elements to discover such musical patterns with the erosion? The results developed in this article prove that the structuring elements have to be the onsets of the patterns to be discovered. Fortunately, in some cases, onsets are more straightforwardly described than patterns. For example, when patterns are repeated at regular time intervals, onsets are defined by isolated points regularly spaced, and are therefore described using two parameters (the number of points of the onsets and the distance separating each point). In this article, we demonstrate with theorems and lemmas the possibility of discovering patterns using morphological erosion based on such onsets, offering an original method to the discovery of specific patterns that repeat at regular time interval.

4. Discovering repeated patterns using the morphological erosion operation

In this section, we introduce the problem of discovering a musical pattern from its onsets. We then propose a theorem that demonstrates that a musical pattern can be discovered from the erosion by its onsets if it covers the data and is fully compact, meaning that a point of musical data cannot be added to its convex hull, which is a local topological condition. We then reveal that the patterns discovered by an erosion are exactly the MTEC patterns, in the sense that it is not possible to add points to them, either inside or outside their convex hulls, without decreasing their number of occurrences, which is a global topological condition. Finally, we provide a musical interpretation of these mathematical results before demonstrating how they can be used to generalize point-set algorithm approaches. This section extends our preliminary work (Lascabettes and Bloch 2024) which is mainly in Section 4.2. However, the main difference here is that the definition of being fully compact concerns only one occurrence of the pattern, and we require the pattern to be a subset of the data, i.e. $P \subseteq X$, with the fully compact or MTEC property.

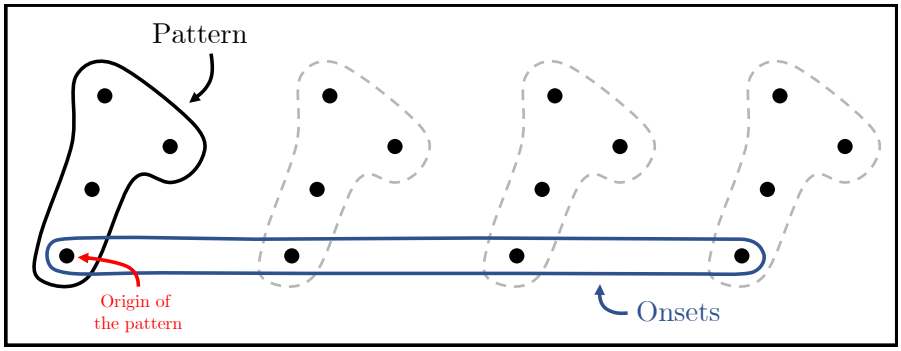
4.1. *Formalization of the problem of discovering a musical pattern from its onsets*

We present here the problem of *discovering a musical pattern from its onsets*. We define the onsets of a pattern as the the set of points that correspond to the translations of the pattern to obtain its occurrences in musical data. This makes the connection with the morphological erosion.

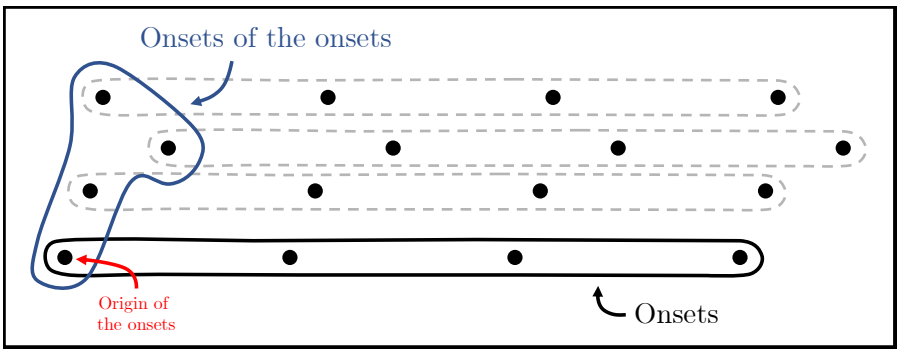
Definition 4.1 (Onsets) Let $P, X \subseteq \mathbb{R}^n$. The *onsets* O of P in X are defined as:

$$O = \varepsilon_P(X) \tag{8}$$

The onsets of a pattern can be interpreted as the lexicographically earliest point of pattern occurrences in musical data. For example, in Figure 2, musical data containing a repeating pattern are represented by dots in a two-dimensional space. In this two-dimensional representation, which is used in the rest of this article, the vertical axis refers to note pitch, while the horizontal axis indicates time. Note that the figures are 2D examples, for the sake of clarity, but all theoretical results apply for any finite dimension. In Figure 2(a), repetitions of the pattern are indicated by dotted lines. The onsets of this pattern include the lexicographically earliest point in each of the four occurrences of the pattern. The interesting result is illustrated in Figure 2(b) by now considering the onsets as a pattern: the onsets of the pattern onsets are equal to the pattern in this particular example. In other words, by computing the points where the onsets occur, we discover the pattern (the occurrences of the set of onsets are indicated by the dotted lines in the figure). This result is quite surprising, and we can formalize it with morphological operators by the problem of discovering a musical pattern from its onsets.



(a) Representation of the pattern and its onsets, which are interpreted as the points where the pattern begins. Pattern repeats are indicated by the dotted-line border. The origin of the pattern is chosen at the starting point of the pattern (assuming that the time axis is from left to right).



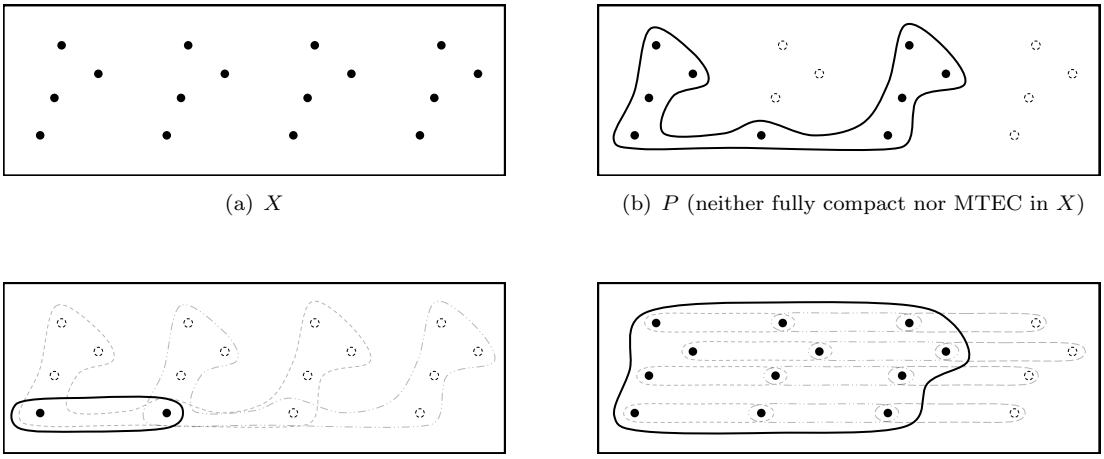
(b) The onsets of the onsets are equal to the pattern. Onsets repeats are indicated by the dotted-line border. That is, the dots corresponding to where the onsets begin are the same as the pattern dots. As before, the origin of the onsets is chosen at its starting point.

Figure 2. Presentation of the problem of discovering the pattern from its onsets. In this example, the onsets of the onsets of the pattern are equal to the pattern, which shows that it is sufficient to know the onsets to discover the pattern.

Definition 4.2 (Discovering a musical pattern from its onsets) Let $P, O, X \subseteq \mathbb{R}^n$, where $O = \varepsilon_P(X)$. The problem of *discovering a musical pattern from its onsets* is to understand when the following equation is true:

$$P = \varepsilon_O(X) \tag{9}$$

Let $P, O, X \subseteq \mathbb{R}^n$ such that $O = \varepsilon_P(X)$. This can be interpreted as X representing the musical data, P a musical pattern and O its onsets. First of all, the problem of discovering a musical pattern from its onsets, i.e. the equality $P = \varepsilon_O(X)$, is not always true as shown in Figure 3. However, Lemma 4.3 allows us to state that the inclusion $P \subseteq \varepsilon_O(X)$ is always true. Therefore, P is always included in the onsets of O. This implies that taking the onsets of the onsets of a pattern may enlarge it, as can be seen in Figure 3 where P, represented in Figure 3(b), is included in $\varepsilon_O(X)$, represented in Figure 3(d).



(a) X

(b) P (neither fully compact nor MTEC in X)

(c) $O = \varepsilon_P(X)$

(d) $P \neq \varepsilon_O(X)$

Figure 3. Example of a pattern, P, that is neither fully compact nor MTEC in X and therefore $P \neq \varepsilon_O(X)$. However the inclusion $P \subseteq \varepsilon_O(X)$ is always satisfied (the origin is the first point on the left).

Lemma 4.3 *(Pattern extension) Let $P, O, X \subseteq \mathbb{R}^n$, such that $O = \varepsilon_P(X)$ and $O \neq \emptyset$. The following inclusion is always true:*

$$P \subseteq \varepsilon_O(X) \tag{10}$$

Proof of Lemma 4.3. Under the assumptions of Lemma 4.3, $O = \varepsilon_P(X)$ is non-empty, we therefore have the following implications.

$$\begin{array}{lll} & \forall o \in O, & P_o \subseteq X \\ \Rightarrow & \forall o \in O, \forall p \in P, & p + o \in X \\ \Rightarrow & \forall p \in P, \forall o \in O, & p + o \in X \\ \Rightarrow & \forall p \in P, & O_p \subseteq X \\ \Rightarrow & \forall p \in P, & p \in \varepsilon_O(X) \end{array}$$

■

In the remainder of the section, we provide additional relations between P and O using morphological operators and we prove that the problem of discovering a musical pattern from its onsets can be solved under some assumptions that are musically interpretable. In particular, we approach this problem in two different ways. First, in Section 4.2, we demonstrate that patterns can be discovered from their onsets if they satisfy a local topological property, i.e. a property that concerns the inside of the patterns. Then, in Section 4.3, we completely solve the problem of discovering a musical pattern from its onsets. We prove that a pattern is discovered from its onsets if and only if it is MTEC in the sense that no points can be added to it without reducing its number of occurrences, which is a property that is not local because it depends on the nature of the entire dataset. These two results represent a new approach to the discovery of patterns in a multidimensional representation of music.

4.2. *The fully compact property: A local sufficient condition for discovering patterns by their onsets*

By setting $P, O, X \subseteq \mathbb{R}^n$, such that $P \subseteq X$, and $O = \varepsilon_P(X)$, the equality $P = \varepsilon_O(X)$ is not always true so we need to add an additional condition on P and X to ensure this equality. In this section, we propose to introduce a local topological condition to the pattern in order to discover it from its onsets, i.e. a property that relies on the inside of the pattern and does not depend on its outside. First, we introduce the definition of being *fully compact*, illustrated in Figure 4, that is, to some extent, the counterpart of the connectivity (without holes) used in image processing. This definition is related to the *compactness* of a pattern, which is the ratio between the number of points of the pattern and the number of points in the dataset within the region spanned by the pattern which could be, for example, the convex hull (Meredith, Lemström, and Wiggins 2002). Moreover, in the OMNISIA implementation of COSIATEC and the other SIA algorithms, the fully compact property is embodied in *TEC compactness*[2] and *occurrence compactness*, and both are controllable by software parameters (Meredith 2016). Let us note $\mathrm{CH}(P)$ the *convex hull* of P defined by $\mathrm{CH}(P) = \{\sum_{i=1}^{k} \lambda_i p_i \mid k \in \mathbb{N}^* \wedge \sum_{i=1}^{k} \lambda_i = 1 \wedge \forall i \in [\![1, k]\!], p_i \in P, \lambda_i \geq 0\}$, where $[\![1, k]\!]$ is the set of integers from 1 to k included.

Definition 4.4 (P fully compact in X) Let $P, X \subseteq \mathbb{R}^n$. P is *fully compact* in X if $\mathrm{CH}(P)$ does not contain any point of X that is not in P, i.e. if:

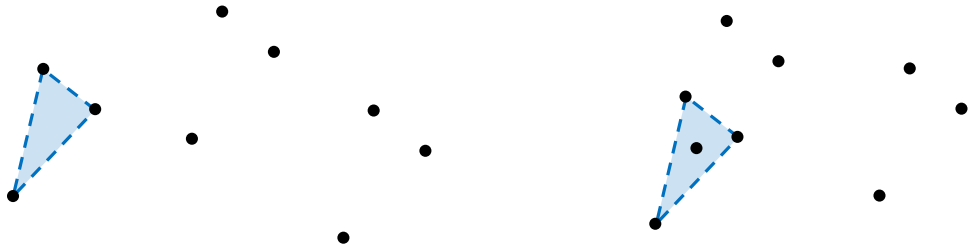
$$\mathrm{CH}(P) \cap X = P \tag{11}$$

Notice that P is fully compact in X implies that $P \subseteq X$. We also assume the particular case where the data X is composed of patterns P which are repeated with translations, that is to say $\gamma_P(X) = X$. Under the condition that P covers X and is fully compact, Theorem 4.5 provides the equality $P = \varepsilon_O(X)$. Therefore, it is possible, under the assumptions of the theorem, to discover the pattern P from its onsets O.

Theorem 4.5 (Fully compact patterns are discovered by their onsets) *Let $P, O, X \subseteq \mathbb{R}^n$ such that $O = \varepsilon_P(X)$, $\gamma_P(X) = X$, P is fully compact in X, and X is bounded. Then it holds that:*

$$P = \varepsilon_O(X) \tag{12}$$

[2]The definition of TEC is provided later in the article (see Definition 4.9).



(a) P is fully compact in X (b) P is not fully compact in X

Figure 4. Musical pattern P composed of three points of X (set of dots) and its convex hull represented in dotted line. P is fully compact in X in (a) because there are no dots inside its convex hull. However, this is not the case for (b).

We provide another link between P and O using morphological operators with the following lemma, which is useful for the proof of Theorem 4.5.

Lemma 4.6 (Closing a pattern by its onsets) *Let $P, O, X \subseteq \mathbb{R}^n$, such that $O = \varepsilon_P(X)$ and $\gamma_P(X) = X$. Then:*

$$\varepsilon_O(X) = \varphi_O(P) \tag{13}$$

Proof of Lemma 4.6. Under the assumptions of the lemma, we have the following implications.

$$\begin{array}{lll} & \gamma_P(X) = X & \\ \Rightarrow & \delta_P(\varepsilon_P(X)) = X & \text{(definition of } \gamma_P) \\ \Rightarrow & \delta_P(O) = X & \text{(definition of } O) \\ \Rightarrow & \delta_O(P) = X & \text{(commutativity of } \oplus) \\ \Rightarrow & \varepsilon_O(\delta_O(P)) = \varepsilon_O(X) & \text{(composing by } \varepsilon_O) \\ \Rightarrow & \varphi_O(P) = \varepsilon_O(X) & \text{(definition of } \varphi_O) \end{array}$$

■

Also, Lemma 4.7, which is useful for the proof of Theorem 4.5, reveals that if P is included in X, then the erosion of X by O is always included in X.

Lemma 4.7 (Inclusion of the onsets erosion in X) *Let $P, O, X \subseteq \mathbb{R}^n$, such that $P \subseteq X$ and $O = \varepsilon_P(X)$. Then:*

$$\varepsilon_O(X) \subseteq X \tag{14}$$

Proof of Lemma 4.7. Under the assumptions of the lemma, we have the following implications using the anti-extensivity property of erosion.

$$P \subseteq X \;\Rightarrow\; 0_{\mathbb{R}^n} \in \varepsilon_P(X) = O \;\Rightarrow\; \varepsilon_O(X) \subseteq X$$

■

Proof of Theorem 4.5. Under the assumptions of the theorem, we can use Lemma 4.6, Lemma 4.3 and Lemma 4.7. We prove that $P = \varepsilon_O(X)$ with two inclusions.

$\boxed{\subseteq}$ The first inclusion comes directly from Lemma 4.3.

$\boxed{\supseteq}$ It has been proved by Serra that for every $C \subseteq \mathbb{R}^n$ convex and $S \subseteq \mathbb{R}^n$ bounded, we have: $\varphi_S(C) = C$ (proposition IV-4 (Serra 1982)). Since X is bounded, $O = \varepsilon_P(X)$ is also bounded and $\mathrm{CH}(P)$ is convex by definition. Consequently, we have:

$$\varphi_O(\mathrm{CH}(P)) = \mathrm{CH}(P) \tag{15}$$

Therefore, we have the following implications.

$$\begin{array}{lll}
 & P \subseteq \mathrm{CH}(P) & \\
\Rightarrow & \varphi_O(P) \subseteq \varphi_O(\mathrm{CH}(P)) & (\varphi_O \text{ is increasing}) \\
\Rightarrow & \varphi_O(P) \subseteq \mathrm{CH}(P) & (\varphi_O(\mathrm{CH}(P)) = \mathrm{CH}(P)) \\
\Rightarrow & \varepsilon_O(X) \subseteq \mathrm{CH}(P) & (\varphi_O(P) = \varepsilon_O(X) \text{ Lemma 4.6}) \\
\Rightarrow & \varepsilon_O(X) \cap X \subseteq \mathrm{CH}(P) \cap X & (\text{intersection with } X) \\
\Rightarrow & \varepsilon_O(X) \subseteq \mathrm{CH}(P) \cap X & (\varepsilon_O(X) \cap X = \varepsilon_O(X) \text{ Lemma 4.7}) \\
\Rightarrow & \varepsilon_O(X) \subseteq P & (\mathrm{CH}(P) \cap X = P \text{ because } P \text{ is fully compact})
\end{array}$$

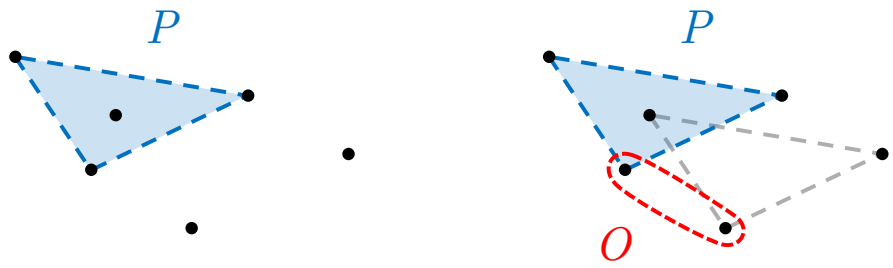
■

Remark 4.8 (On the importance of the assumptions of Theorem 4.5)

- P is fully compact in X. Figure 3 illustrates a pattern P in X that is not fully compact but satisfies the other assumptions of Theorem 4.5 with $\gamma_P(X) = X$ and X bounded. In this case, we have $\varepsilon_O(X) \neq P$ which demonstrates the importance of the property of being fully compact.
- $\gamma_P(X) = X$. Figure 6 illustrates a pattern P in X such that $\gamma_P(X) \neq X$ but satisfies the other assumptions of Theorem 4.5 with P fully compact in X and X bounded. In this case, we have $\varepsilon_O(X) \neq P$, which demonstrates the importance of X being composed solely of P patterns.

4.3. *The MTEC property: A global necessary and sufficient condition for discovering patterns by their onsets*

In the previous section, we proved that a pattern P can be discovered from its onsets O with a morphological erosion, i.e. $P = \varepsilon_O(X)$, if it is fully compact, covers X and X is bounded. However, these are not necessary and sufficient conditions, i.e. there are patterns that do not satisfy these assumptions and still verify the equality $P = \varepsilon_O(X)$. For example, in Figure 5, the pattern P is not fully compact (Figure 5(a)), but the equality $P = \varepsilon_O(X)$ is true (Figure 5(b)). So which patterns exactly satisfy $P = \varepsilon_O(X)$ when $O = \varepsilon_P(X)$ and X is not necessarily equal to $\gamma_P(X) = X$? In this section, we prove that these are exactly the MTEC patterns, in the sense that no points can be added to them without reducing the frequency with which they occur in X. The difference with the fully compact property is that the added points can be either inside or outside the pattern's convex hull. This MTEC property ensures the equivalence with the equality $P = \varepsilon_O(X)$. In other words, we prove that the problem of discovering a musical pattern from its onsets is equivalent to obtain MTEC patterns.



(a) P is not fully compact in X (b) P is MTEC

Figure 5. Example of a musical pattern P which is not fully compact in X (set of dots) but can be discovered by its onsets O because $P = \varepsilon_O(X)$. Consequently, (P, O) form an MTEC conjugate pair.

We start by defining the *translational equivalence class* (TEC) of a pattern P by the set of patterns from X that are equal to P up to a translation (Meredith, Lemström, and Wiggins 2002).

Definition 4.9 (TEC) Let $P, X \subseteq \mathbb{R}^n$ such that $P \subseteq X$. The *translational equivalence class* (TEC) for P in X is defined as follows:

$$TEC(P, X) = \{P_t \mid \exists t \in \mathbb{R}^n \wedge P_t \subseteq X\} \tag{16}$$

From this definition, the MTEC property describes a category of repeated patterns that are maximal with respect to their TEC, called *maximal translational equivalence class* (MTEC) introduced by Collins and Meredith (2013).

Definition 4.10 (MTEC) Let $P, X \subseteq \mathbb{R}^n$ such that $P \subseteq X$. P is defined as *maximal with respect to its translational equivalence class* (MTEC) if:

$$\forall P' \subseteq \mathbb{R}^n, P \subsetneq P' \Rightarrow |TEC(P', X)| < |TEC(P, X)| \tag{17}$$

In other words, if the pattern is enriched with new points, then this decreases its number of occurrences within the dataset. For example, in Figure 6, the pattern P is not MTEC because there are three occurrences of P in X and it is possible to add the point on the left and still have three occurrences of the pattern in the dataset. By adding the point on the left, the pattern becomes MTEC.

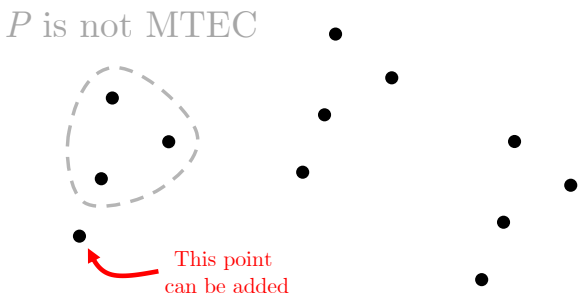


Figure 6. Illustration of a pattern, P , that is not MTEC: it is possible to add the point on the left without decreasing the frequency with which the pattern occurs in X.

With this definition, we propose to completely solve the problem of discovering a musical pattern from its onsets with Theorem 4.11.

THEOREM 4.11 (MTEC patterns are those discovered by their onsets) *Let $P, O, X \subseteq \mathbb{R}^n$ such that $P \subseteq X$ and $O = \varepsilon_P(X)$. The following equivalence holds:*

$$P \text{ is MTEC} \iff P = \varepsilon_O(X) \tag{18}$$

Proof of Theorem 4.11. Under the assumptions of the theorem, we prove it by double implications.

$\boxed{\Leftarrow}$ We prove this implication by contraposition, i.e. we have to prove the following.

$$P \text{ is not MTEC } \Rightarrow P \neq \varepsilon_O(X) \tag{19}$$

Therefore, suppose that P is not MTEC. So there exists $P' \subseteq \mathbb{R}^n$ such that $P \subsetneq P'$ and $|\varepsilon_{P'}(X)| \geq |\varepsilon_P(X)|$. We set that "decreasingness of ε" signifies decreasingness of the erosion with respect to the structuring element. Then:

$$\begin{aligned}
& P \subseteq P' & \\
\Rightarrow\ & \varepsilon_{P'}(X) \subseteq \varepsilon_P(X) & \text{(decreasingness of } \varepsilon) \\
\Rightarrow\ & \varepsilon_{P'}(X) = \varepsilon_P(X) & (|\varepsilon_{P'}(X)| \geq |\varepsilon_P(X)|)
\end{aligned}$$

The following implications prove the first part of the theorem:

$$\begin{aligned}
& P' \setminus P \subseteq P' & \\
\Rightarrow\ & \varepsilon_{P'}(X) \subseteq \varepsilon_{P' \setminus P}(X) & \text{(decreasingness of } \varepsilon) \\
\Rightarrow\ & \varepsilon_P(X) \subseteq \varepsilon_{P' \setminus P}(X) & (\varepsilon_{P'}(X) = \varepsilon_P(X)) \\
\Rightarrow\ & O \subseteq \varepsilon_{P' \setminus P}(X) & (O = \varepsilon_P(X)) \\
\Rightarrow\ & \varepsilon_{\varepsilon_{P' \setminus P}(X)}(X) \subseteq \varepsilon_O(X) & \text{(decreasingness of } \varepsilon) \\
\Rightarrow\ & P' \setminus P \subseteq \varepsilon_O(X) & (P' \setminus P \subseteq \varepsilon_{\varepsilon_{P' \setminus P}(X)}(X) \text{ Lemma 4.3}) \\
\Rightarrow\ & P \neq \varepsilon_O(X) & (\exists p' \in P' : p' \notin P \wedge p' \in \varepsilon_O(X) \text{ because } P \neq P')
\end{aligned}$$

$\boxed{\Rightarrow}$ We reason again by contraposition, i.e. we have to prove the following.

$$P \neq \varepsilon_O(X) \Rightarrow P \text{ is not MTEC} \tag{20}$$

Thus, suppose that: $P \neq \varepsilon_O(X)$. Because Lemma 4.3 implies that $P \subseteq \varepsilon_O(X)$, we deduce $P \subsetneq \varepsilon_O(X)$. Therefore, by setting $P' = \varepsilon_O(X)$, we then have $P \subsetneq P'$. Applying Lemma 4.3 to $\varepsilon_P(X)$ yields the following implications.

$$\begin{aligned}
& \varepsilon_P(X) \subseteq \varepsilon_{\varepsilon_{\varepsilon_P(X)}(X)}(X) \\
\Rightarrow\ & \varepsilon_P(X) \subseteq \varepsilon_{\varepsilon_O(X)}(X) \\
\Rightarrow\ & \varepsilon_P(X) \subseteq \varepsilon_{P'}(X) \\
\Rightarrow\ & |\varepsilon_P(X)| \leq |\varepsilon_{P'}(X)|
\end{aligned}$$

Therefore, we conclude that P is not MTEC because there exists $P' = \varepsilon_O(X)$ such that $P \subsetneq P'$ and $|\varepsilon_{P'}(X)| \geq |\varepsilon_P(X)|$. ■

In addition to solving the problem of discovering a musical pattern from its onsets, Theorem 4.11 provides another result about the nature of patterns discovered from an erosion. Indeed, while Lemma 3.1 has shown that the morphological erosion by a bipoint $\{0, v\}$ discovers MTP patterns, Corollary 4.12 reveals that the erosion by any structuring element discovers MTEC patterns. This reveals that morphological erosion generalizes the way musical patterns are discovered, not by using a structuring element composed of 0 and a vector, but by the possibility of using any structuring element. Morphological erosion thus proves to be a fundamental tool in the musical pattern discovery task.

Corollary 4.12 (Characterization of MTEC patterns with the erosion) *Let $P, X \subseteq \mathbb{R}^n$ such that $P \subseteq X$. The following equivalence holds:*

$$P \text{ is MTEC} \iff \exists S \subseteq \mathbb{R}^n : P = \varepsilon_S(X) \tag{21}$$

Proof of Corollary 4.12. Let $P, O, X \subseteq \mathbb{R}^n$ such that $P \subseteq X$ and $O = \varepsilon_P(X)$.
$\boxed{\Rightarrow}$ Suppose that P is MTEC, using Theorem 4.11 this means that $P = \varepsilon_O(X)$. By setting $S = O$, we therefore have the existence of $S \subseteq \mathbb{R}^n$ such that: $P = \varepsilon_S(X)$.
$\boxed{\Leftarrow}$ Let $S \subseteq \mathbb{R}^n$ such that $P = \varepsilon_S(X)$.

$$\begin{array}{lll} & S \subseteq \varepsilon_{\varepsilon_S(X)}(X) & \text{(Lemma 4.3 applied to } S) \\ \Rightarrow & S \subseteq \varepsilon_P(X) & (P = \varepsilon_S(X)) \\ \Rightarrow & S \subseteq O & (O = \varepsilon_P(X)) \\ \Rightarrow & \varepsilon_O(X) \subseteq \varepsilon_S(X) & \text{(decreasingness of } \varepsilon) \\ \Rightarrow & \varepsilon_O(X) \subseteq P & (P = \varepsilon_S(X)) \end{array}$$

Lemma 4.3 applied to P leads to the other inclusion: $P \subseteq \varepsilon_O(X)$. Therefore $P = \varepsilon_O(X)$, which proves that P is MTEC using Theorem 4.11. ■

This result offers an original method for discovering some MTEC patterns. In particular, it reveals that morphological erosion by any structuring element produces an MTEC pattern. This might be a solution to a problem posed by T. Collins and D. Meredith when they stated that:

> "We need to gain a better understanding of how to discover MTEC patterns." (Collins and Meredith 2013)

Indeed, these results may provide useful insights for the design of an efficient algorithm (not with exponential time complexity) to discover all the MTEC patterns in a dataset, or at least the most interesting ones.

4.4. *Discover repeated patterns at regular intervals using morphological erosion*

One of the most important results of this article is the possibility of discovering musical patterns using morphological erosion by the onsets under the assumptions mentioned in Theorem 4.5 or Theorem 4.11. Figure 7 shows a simple example to illustrate these two theorems with the morphological links between the data X, the pattern P, and its onsets O. This figure displays Theorem 4.5 because it is composed of a repeated pattern which is fully compact, and Theorem 4.11 because the pattern is MTEC. Therefore, these results can be used to discover *exact repetitive patterns* by considering structuring

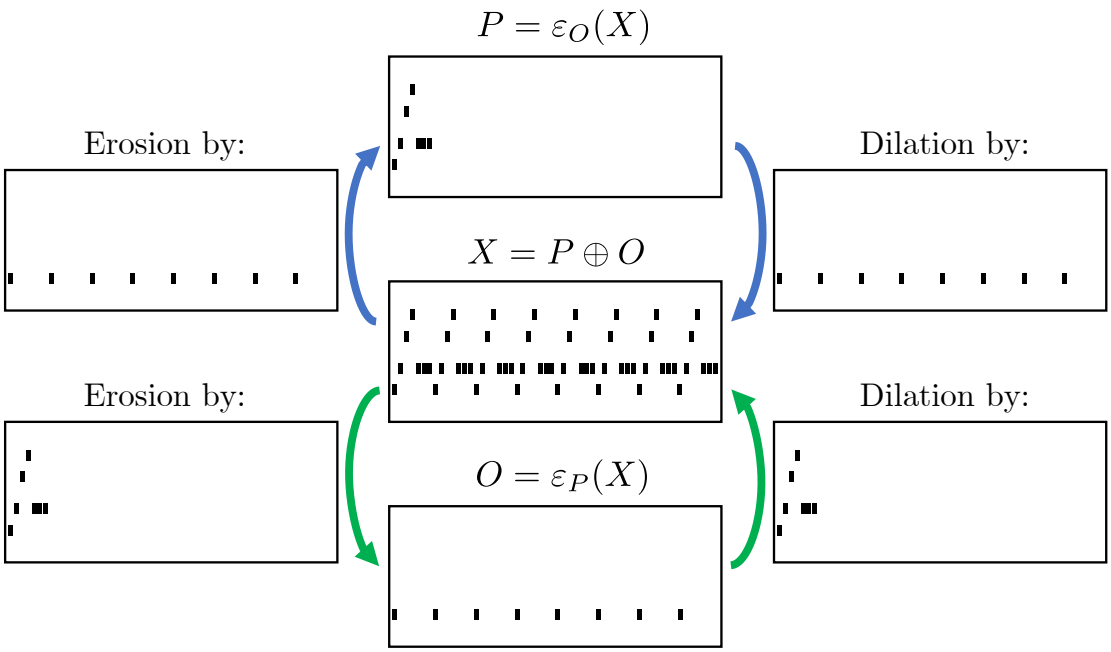


Figure 7. Illustration using morphological operators of the relations between the data X, the musical pattern P, and its onsets O. In this case, the pattern P can be discovered from morphological erosion because it is fully compact in X or is also MTEC, and therefore satisfies the assumptions of Theorem 4.5 and Theorem 4.11.

elements that represent time-periodic onsets. For example, if we are looking for musical patterns P that repeat in time at regular intervals of length $L > 0$, the structuring element O associated with the onsets is as follows (where the first coordinate iL represents the temporal component):

$$\{(iL, 0, \ldots, 0) \mid i \in [\![0, m-1]\!]\} \tag{22}$$

In this case, we are looking for at least m exact repetitions of the musical pattern P. As a result, only one parameter L is required to discover patterns that repeat exactly at regular intervals. This offers an original method for discovering patterns that repeat at regular intervals using morphological erosion by a particular structuring element.

5. MTEC conjugate pair of (pattern, onsets)

In this section, we focus on MTEC conjugate pairs, i.e. pairs of musical patterns and their onsets which can be deduced from each other by using the morphological erosion. We propose a theorem that ensures that, given a pattern, there is always a unique MTEC conjugate pair that has the same onsets, that is to say, applying twice the erosion by a pattern produces the smallest MTEC pattern containing it in the sense of inclusion.

5.1. *Definition of an MTEC conjugate pair*

In the following, we define the musical patterns P and their onsets O which verify the equality $P = \varepsilon_O(X)$ as *MTEC conjugate patterns*, and the pair (P, O) is an *MTEC conjugate pair*.

Definition 5.1 (MTEC conjugate pair) Let $P, O, X \subseteq \mathbb{R}^n$. The pair (P, O) is an *MTEC conjugate pair* if it satisfies the following two equalities:

$$P = \varepsilon_O(X) \quad \text{and} \quad O = \varepsilon_P(X) \tag{23}$$

An example of an MTEC conjugate pair is displayed in Figure 8.

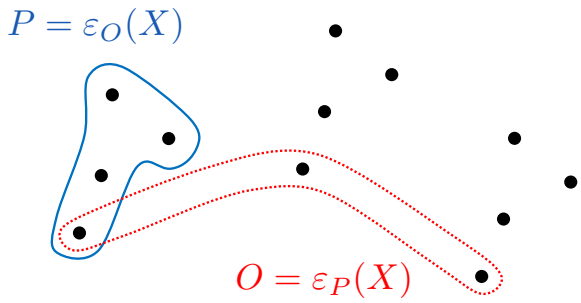


Figure 8. Illustration of an MTEC conjugate pair (P, O): we have $P = \varepsilon_O(X)$ and $O = \varepsilon_P(X)$.

5.2. *Existence of MTEC conjugate pairs*

Since it is not always true that $O = \varepsilon_P(X)$ and $P = \varepsilon_O(X)$, it is in principle not always possible to discover an MTEC conjugate pair (P, O) of a musical pattern and its onsets. Therefore, given a musical pattern P, it cannot always be obtained from its onsets O with an erosion because P may not be MTEC. This can be confirmed because several different patterns can have the same onsets. But the erosion by these onsets can only be equal to one of these patterns at most. With this in mind, Theorem 5.2 assures that given a pattern P, there exists an MTEC conjugate pair (P', O) such that the onsets of P' are the same as the onsets of P.

THEOREM 5.2 (MTEC conjugate with the onsets) *Let $P, O, X \subseteq \mathbb{R}^n$, such that $O = \varepsilon_P(X)$. By defining $P' = \varepsilon_O(X)$, we have:*

$$O = \varepsilon_{P'}(X) \tag{24}$$

and therefore (P', O) is an MTEC conjugate pair.

Proof of Theorem 5.2. We prove the lemma with two inclusions.
$\boxed{\subseteq}$ The first inclusion comes from Lemma 4.3 by changing the role of P and O by O and P'.
$\boxed{\supseteq}$ Using Lemma 4.3 we obtain: $P \subseteq \varepsilon_O(X)$. The erosion is decreasing according to the structuring element, hence:

$$P \subseteq \varepsilon_O(X) \Rightarrow \varepsilon_{\varepsilon_O(X)}(X) \subseteq \varepsilon_P(X)$$

Since $O = \varepsilon_P(X)$ and $P' = \varepsilon_O(X)$ by definition, this implies $\varepsilon_{P'}(X) \subseteq O$ which proves the other inclusion. ■

Therefore, Theorem 5.2 can be reformulated with Corollary 5.3, which states that given the onsets O from a pattern P, there always exists a unique pattern P' containing P such that P' and O are MTEC conjugate patterns.

Corollary 5.3 (Existence of a unique MTEC conjugate pattern) *Let $P, O, X \subseteq \mathbb{R}^n$, such that $O = \varepsilon_P(X)$. Then:*

$$\exists! P' \subseteq \mathbb{R}^n : P \subseteq P' \wedge P' \text{ and } O \text{ are MTEC conjugate patterns} \tag{25}$$

Proof of Corollary 5.3. Let $P' = \varepsilon_{\varepsilon_P(X)}(X)$, we have $P \subseteq P'$ (Lemma 4.3), P' and O are MTEC conjugate patterns (Theorem 5.2), and uniqueness results from O having at most one MTEC conjugate. ■

Finally, these results also reveal that applying twice the erosion by a pattern results in the smallest MTEC pattern (in the sense of inclusion) that contains it, which is described in Corollary 5.4. This result is relevant for MTEC pattern discovery, as it proposes a method for obtaining the smallest MTEC patterns from a set of patterns.

Corollary 5.4 (Smallest MTEC pattern) *Let $P, X \subseteq \mathbb{R}^n$ such that $P \subseteq X$. Then $\varepsilon_{\varepsilon_P(X)}(X)$ is the smallest MTEC pattern in the sense of inclusion that contains P:*

$$\nexists Q \subseteq \mathbb{R}^n \text{ which is MTEC } : P \subsetneq Q \subsetneq \varepsilon_{\varepsilon_P(X)}(X) \tag{26}$$

Proof of Corollary 5.4. Let $P, Q, X \subseteq \mathbb{R}^n$ such that $P \subseteq X$. First, because $P \subseteq X$, Lemma 4.7 satisfies the condition that $\varepsilon_{\varepsilon_P(X)}(X) \subseteq X$. We assume by contradiction that Q is MTEC and that $P \subseteq Q \subseteq \varepsilon_{\varepsilon_P(X)}(X)$.

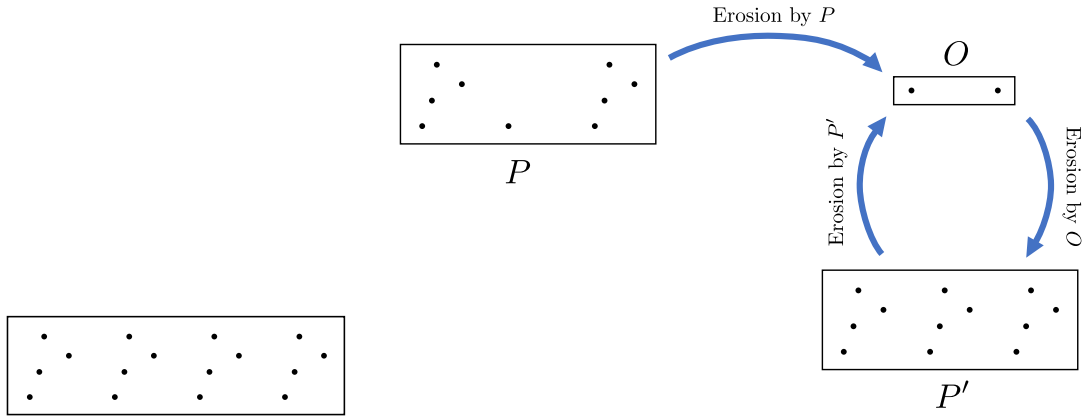
$$\begin{array}{lll}
 & P \subseteq Q & \\
\Rightarrow & \varepsilon_Q(X) \subseteq \varepsilon_P(X) & \text{(decreasingness of } \varepsilon) \\
\Rightarrow & \varepsilon_{\varepsilon_P(X)}(X) \subseteq \varepsilon_{\varepsilon_Q(X)}(X) & \text{(decreasingness of } \varepsilon) \\
\Rightarrow & \varepsilon_{\varepsilon_P(X)}(X) \subseteq Q & \text{(Theorem 4.11 with } Q \text{ MTEC)} \\
\Rightarrow & \varepsilon_{\varepsilon_P(X)}(X) = Q & \text{(By assumption } Q \subseteq \varepsilon_{\varepsilon_P(X)}(X))
\end{array}$$

This reasoning proves that Q cannot be different from $\varepsilon_{\varepsilon_P(X)}(X)$ and proves the corollary. ■

5.3. *Musical interpretations of an MTEC conjugate pair*

In Figure 3, the pattern, P, and its onsets, O, do not form an MTEC conjugate pair because $P \neq \varepsilon_O(X)$. However, according to Theorem 5.2, by defining $P' = \varepsilon_O(X)$, we obtain an MTEC conjugate pair, (P', O), where O is the onsets of P and P' is the pattern P with additional points for it to be MTEC. This is illustrated in Figure 9 where the morphological erosion yields the MTEC conjugate pair (P', O) from P, and we can confirm that $P \subseteq P'$.

Remark 5.5 (Relation between the pattern P and its onsets O) Under the assumptions of Theorem 4.5 or Theorem 4.11, we can switch from P to O with the erosion.



(a) X

(b) Relations between P, O and P'

Figure 9. Using the example of Figure 3, (P, O) is not an MTEC conjugate pair because $P \neq \varepsilon_O(X)$. However, the morphological erosion by O provides the pattern P', by adding points to P, such that (P', O) is an MTEC conjugate pair because $P' = \varepsilon_O(X)$ and $\varepsilon_{P'}(X) = O$.

$$\begin{tikzcd} P \arrow[r, bend left, "\varepsilon_P(X)"] & O \arrow[l, bend left, "\varepsilon_O(X)"] \end{tikzcd}$$

In this case, it is sufficient to know P (resp. O) to determine O (resp. P). However, when the assumptions are not met, Theorem 5.2 states that it is enough to apply one erosion to get into a (pattern, onsets) cycle.

$$\begin{tikzcd} P \arrow[r, "\varepsilon_P(X)"] & O \arrow[r, bend left, "\varepsilon_O(X)"] & P' \arrow[l, bend left, "\varepsilon_{P'}(X)"] \end{tikzcd}$$

Therefore, we can always obtain an MTEC conjugate pair of a pattern and its onsets from any musical pattern P. If it respects some properties, such as those of Theorem 4.5 or Theorem 4.11, this pair is trivial because it consists of the pattern P and its onsets O. Otherwise, we obtain the MTEC conjugate pair by completing the musical pattern with additional points with a morphological erosion as stated in Theorem 5.2.

5.4. *Points covered by the opening of an MTEC conjugate pair*

The following lemma states that if we have an MTEC conjugate pair (P, O), then the points covered by an opening by P are the same as the ones covered by an opening by O. This lemma is very useful in our situation because it demonstrates that the choice of P or O does not have any importance when we are interested in the points covered by an opening.

Lemma 5.6 (Covered points by MTEC conjugate patterns) *Let $P, O, X \subseteq \mathbb{R}^n$, such that $P = \varepsilon_O(X)$ and $O = \varepsilon_P(X)$. Then it is equivalent to perform an opening of X by P or by O:*

$$\gamma_P(X) = \gamma_O(X) \tag{27}$$

Proof of Lemma 5.6. Under the assumptions of the lemma, we directly prove the equality: $\gamma_P(X) = \delta_P(\varepsilon_P(X)) = \delta_P(O) = \delta_O(P) = \delta_O(\varepsilon_O(X)) = \gamma_O(X)$. ■

Lemma 5.6 can be visualized in Figure 2, where the set of points covered by the opening with the pattern as structuring element is the same as the set of points covered by the opening with the onsets.

6. Applications to symbolic representations of music

In this section, we apply the method proposed in this article to discover patterns in musical data. We therefore demonstrate that musical patterns can be discovered from their onsets and show that a musical piece can be approximated as a union of dilations between patterns and onsets.

6.1. *Symbolic representations of music to apply mathematical morphology*

The first thing to consider when applying mathematical morphology operators to music is how to represent the data. As described in Section 3, musical data must belong to $\mathbb{R}^n$. This is well adapted to symbolic representations of music because each note can be defined by a tuple as it has been done before by Meredith, Lemström, and Wiggins (2002) to discover musical patterns. We propose here to work in $\mathbb{R}^3$, where each note is a unique point defined by (o, p, d), where o is the note onset (i.e. when the note appears in time), p is the pitch of the note, and d its duration (it is also possible to add other dimensions such as velocity or voice). This representation ensures that the number of notes of a musical piece X is given by $|X|$. Karvonen and Lemström (2008) have chosen to represent the duration in the piano roll. In this case, each note is no longer a single point but a segment of length proportional to the duration of the note. From a computational point of view, this increases the computation time considerably because $|X|$ is much larger than the number of notes in X, which is why we did not choose to follow this path. However, this idea would be particularly well suited for pattern discovery in music performance because it would allow for a temporal approximation when learning the structuring elements. Therefore, in the rest of this article, musical data belong to $\mathbb{R}^3$ and the morphological operators are applied in $\mathbb{R}^3$. However, to avoid having figures in $\mathbb{R}^3$ that are difficult to interpret, we project the result in a piano roll representation to simplify understanding.

6.2. *Discovering the patterns from the onsets and summarize musical data with morphological operators*

We apply here the proposed method to a musical example in order to discover the main musical patterns and to approximate the musical data with a union of dilations between patterns and onsets. Regarding the musical data X, we choose here the guitar part of the musical piece "Deserted Dunes Welcome Weary Feet" of the band King Gizzard and the Lizard Wizard, illustrated in Figure 10 using a piano roll representation. This song is copyright free and the midi files are freely available[3], which makes it a good candidate for music information retrieval research. This musical example has been chosen to illustrate

[3]The whole album is available as midi files where the tracks are separated at: https://kglw8-bit.bandcamp.com/album/polygondwanaland-8-bit-w-tabs

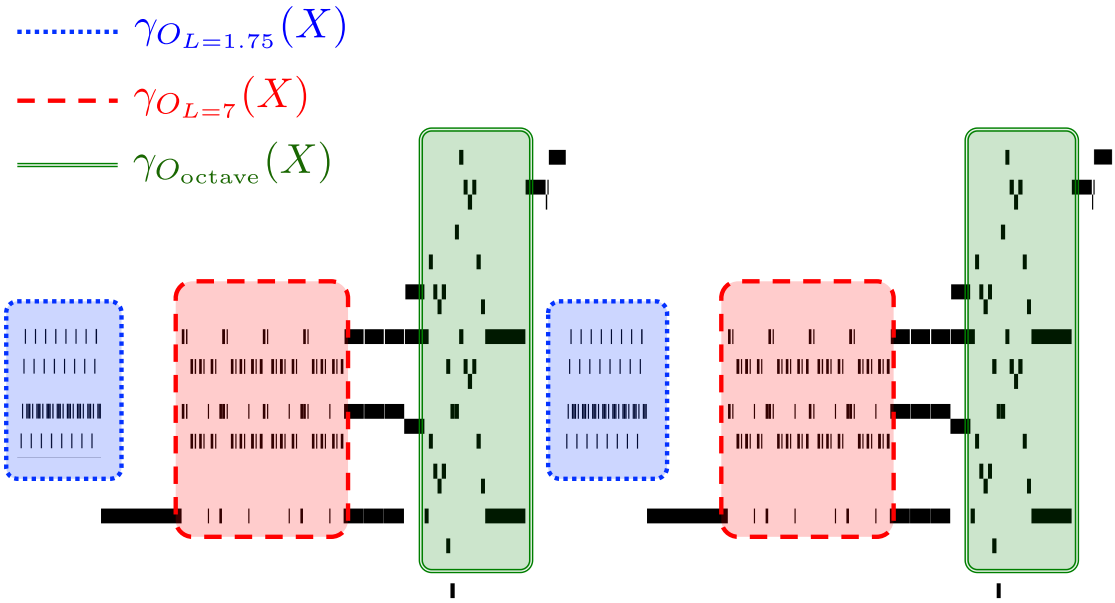


Figure 10. Representation of the musical data X from the midi file of the piece *Deserted Dunes Welcome Weary Feet* (guitar voice). X is composed of 366 notes, $\gamma_{O_{L=1.75}}(X)$, $\gamma_{O_{L=7}}(X)$, and $\gamma_{O_{\text{octave}}}(X)$ discover respectively 112, 144, and 60 notes of X. In all cases, the morphological openings can be decomposed as a dilation of a musical pattern by its onsets.

the results of this article, as it consists almost entirely of patterns that have many consecutive, regularly-spaced occurrences. In this case, X is composed of 366 notes, i.e. $|X| = 366$, and we aim to discover a family of structuring elements $\mathcal{F}$ which represents musical patterns, such that $|\gamma_{\mathcal{F}}(X)|$ is as close as possible to $|X|$. In addition, we choose to discover the MTEC conjugate pairs (P, O) that satisfy :

$$|P \oplus O| = |P| \times |O| \tag{28}$$

In that case, each point $x \in X$ is uniquely expressed as $x = p + o$ with $p \in P$ and $o \in O$. In the case where the equality is not satisfied, we have $|P \oplus O| < |P| \times |O|$ and there is at least a point $x \in X$ such that $x = p + o = p' + o'$ with $p, p' \in P$, $o, o' \in O$ and $p \neq p', o \neq o'$, which means that the information is redundant. Therefore, we define (P, O) as *non-redundant* when Equation 28 is satisfied. In general, it is not always possible to express $P \oplus O$ as the union of non-intersecting occurrences of a pattern, but this is the case for patterns that repeat at regular intervals, which is why we have chosen this criterion. Other criteria have already been mentioned in point-set algorithms for selecting salient pairs from the many discovered ones. In particular, D. Meredith et al. have used the *compression ratio* to select the best pair (P, O) (which is called a TEC in their method) in the COSIATEC algorithm (Meredith, Lemström, and Wiggins 2002; Meredith 2006, 2013). However, in the COSIATEC implementation, the compression ratio must be applied to all pairs, so all pairs need to be discovered to select the best one, whereas with the redundancy criterion, patterns are generated only until a non-redundant pair is found. This is why we have chosen to use the non-redundancy criterion, which is a special case of the compression ratio criterion, as it is sufficient for the discovery of patterns that repeat at regular time intervals.

We can first discover exact repetitions on a large scale, i.e. if sections repeat. To

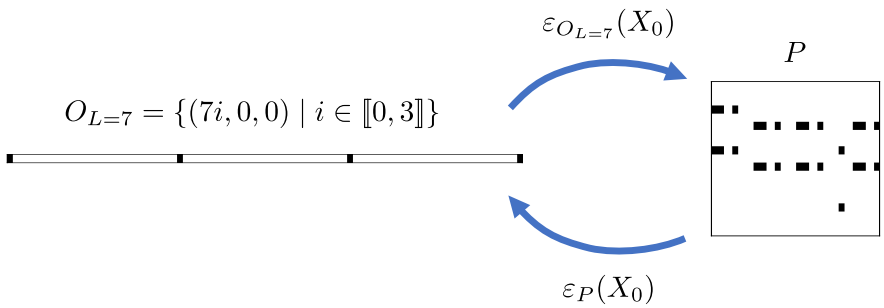
do this, we must look for structuring elements that preserve as many notes as possible after an opening of X in the form $A = \{(0,0,0),(L,0,0)\}$, with L large enough. Let L_{max} be the temporal length of X. We notice that the structuring element $A = \{(0,0,0),(L_{max}/2,0,0)\}$ satisfies: $\gamma_A(X) = X$. This implies that X can be divided into two identical parts with respect to time. Therefore, it is enough to study X only on its first half, i.e. it is enough to study X on $\varepsilon_A(X)$. Since $|\varepsilon_A(X)| = \frac{|X|}{2} = 183$, this allows the data to be reduced considerably. In the following, we define the first half of X as X_0, which is equal to $\varepsilon_A(X)$, to simplify notations and we learn the structuring elements using X_0. A similar reasoning can be used to discover repeated sections, even if X is not divided exactly into two equal parts. In the following, we set the temporal axis so that 1.0 is the duration of a quarter note.

According to Section 4.4, we know the shape of the onsets O if we want to discover musical patterns P that repeat exactly or with transposition. We can then start by discovering the musical patterns P that repeat exactly 4 times in X by looking for the structuring elements (onsets) that are of the following form.

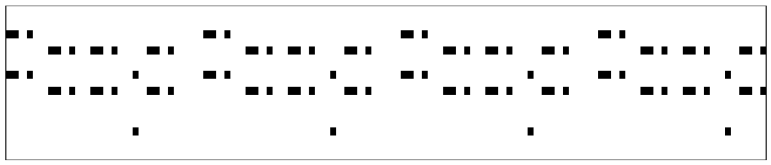
$$O_L = \{(0,0,0),(L,0,0),(2L,0,0),(3L,0,0)\}$$

In this equality, L belongs to $\mathbb{R}$ because the musical data X belong to $\mathbb{R}^3$, so L is not necessarily an integer. By varying L, we can learn the onsets O_L that maximize $|\gamma_{O_L}(X)|$. From the previous paragraph, it is equivalent to maximizing $|\gamma_{O_L}(X_0)|$, i.e. just processing the first half of X. Thus, the value of L that maximizes $|\gamma_{O_L}(X_0)|$ is reached for $L = 7$ with $|\gamma_{O_{L=7}}(X_0)| = 72$. This means that $72 \times 2 = 144$ notes, among the 366 of X, are part of a musical pattern that is repeated with a period of $L = 7$ (the factor 2 in 72×2 comes from the fact that we used X_0 which is the first half of X). According to Theorem 4.5 or Theorem 4.11, we can then use an erosion of X_0 by the structuring element $O_{L=7}$ to discover the corresponding musical pattern P. The onsets $O_{L=7}$ and its associate musical pattern $P = \varepsilon_{O_{L=7}}(X_0)$ are represented in Figure 11(a) using a piano roll representation. Because $|P| \times |O_{L=7}| = |\gamma_{O_{L=7}}(X_0)|$ we can conclude that $(P, O_{L=7})$ is a non-redundant pair, based on Equation 28, in the sense that it best interprets the notions of musical pattern and its onsets without redundancy of information. The set $\gamma_{O_{L=7}}(X)$ is represented in red and framed by long dotted lines in Figure 10. We have just discovered the first structuring element $O_{L=7}$ of the family $\mathcal{F}$ which allows us to describe X as closely as possible. To identify other structuring elements using the same process, the set $\gamma_{O_{L=7}}(X_0)$ is removed from X_0 with the *top-hat transform* operation, which is another morphological operator that removes the results of the opening from the initial set: $T_{O_{L=7}}(X_0) = X_0 \setminus \gamma_{O_{L=7}}(X_0)$, and the discovery of other structuring elements is done with $X_1 = T_{O_{L=7}}(X_0)$.

Continuing the process, the value of L that maximizes $|\gamma_{O_L}(X_1)|$ is reached for $L = 1.75$. In this case, we have $|\gamma_{O_{L=1.75}}(X_1)| = 56$, which means that $56 \times 2 = 112$ notes of X are part of a pattern that repeats with a period of $L = 1.75$. The onsets $O_{L=1.75}$ and their corresponding musical pattern P, obtained by the erosion of X_1 by $O_{L=1.75}$, are represented in Figure 12(a) using a piano roll representation. However, since $|O_{L=1.75}| \times |P| = 140 \neq 56$, the pair $(P, O_{L=1.75})$ is not non-redundant. By looking for 8 but not 4 exact repetitions of a musical pattern with $O'_{L=1.75} = \{(1.75i,0,0) \mid i \in [\![0,7]\!]\}$, we get a non-redundant pair because $|O'_{L=1.75}| \times |P'| = 56$, with $P' = \varepsilon_{O'_{L=1.75}}(X_1)$. This non-redundant pair $(P', O'_{L=1.75})$ is shown in Figure 12(b). Unlike P, it can be seen that P' is not periodic and therefore better corresponds to what we expect from a musical pattern. The result obtained with the morphological operation $\gamma_{O_{L=1.75}}(X)$ is shown in Figure 10 in blue and framed by small dotted lines. As before, we remove $\gamma_{O_{L=1.75}}(X_1)$



(a) $(P, O_{L=7})$ pair where the pattern P is discovered using the erosion by $O_{L=7}$



(b) $\gamma_{O_{L=7}}(X_0) = O_{L=7} \oplus P$

Figure 11. Discovery of repeated patterns by learning the onsets with the exact repetition constraint when $L = 7$. Because $|\gamma_{O_{L=7}}(X_0)| = |P| \times |O_{L=7}|$, we conclude that $(P, O_{L=7})$ is a non-redundant pair.

from X_1 by defining $X_2 = T_{O_{L=1.75}}(X_1)$ to discover the next structuring elements.

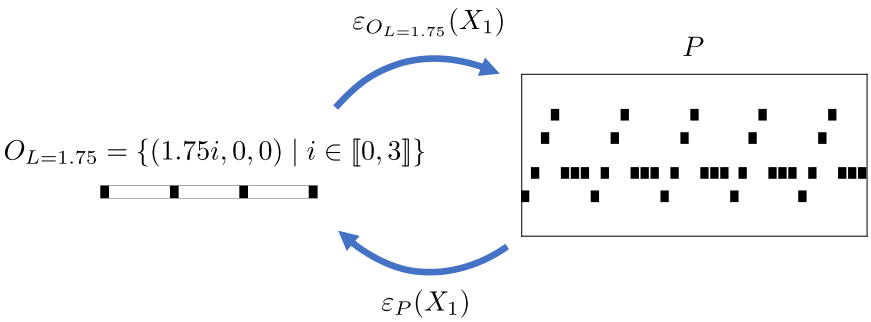
Then, we can also consider repetitions with respect to the pitch. For example, octave intervals can be discovered by the structuring element $O_{\text{octave}} = \{(0, 0, 0), (0, 12, 0)\}$. The first coordinate of the elements of O_{octave} is 0 to discover notes that appear at the same time, and the last coordinate is 0 to discover notes that have the same duration. With this structuring element, we obtain $|\gamma_{O_{\text{octave}}}(X_2)| = 30$. Thus, at least $30 \times 2 = 60$ notes of X are part of an octave interval. By setting $P = \varepsilon_{O_{\text{octave}}}(X_2)$, we obtain the musical pattern that is repeated at the octave. Since $|O_{\text{octave}}| \times |P| = 30$, we can deduce that (O_{octave}, P) is non-redundant. The two structuring elements O_{octave} and P are represented in Figure 13(a). Moreover, the set $\gamma_{O_{\text{octave}}}(X)$ is shown in Figure 10 in green and framed with double lines.

Finally, by setting $\mathcal{F} = \{O_{L=7}, O'_{L=1.75}, O_{\text{octave}}\}$, the set $\gamma_{\mathcal{F}}(X)$ is a fairly good approximation of X in the sense that $|\gamma_{\mathcal{F}}(X)| = 316$. Since $|T_{\mathcal{F}}(X)| = 50$, there are only 50 notes out of 366 that have not been discovered by $\gamma_{\mathcal{F}}(X)$. We can then describe X by morphological operations:

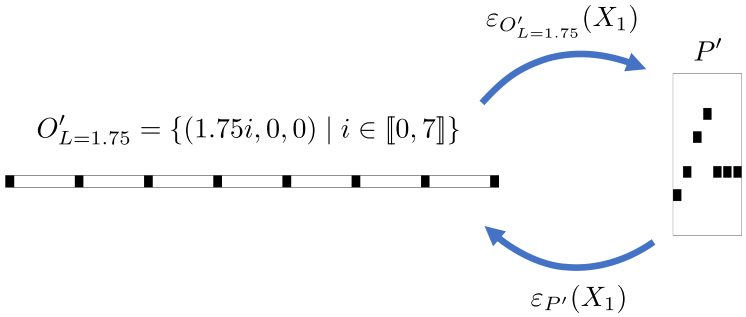
$$X \approx \bigcup_{O \in \mathcal{F}} O \oplus \varepsilon_O(X) \tag{29}$$

6.3. *Other applications: using the method to discover patterns that are not fully compact*

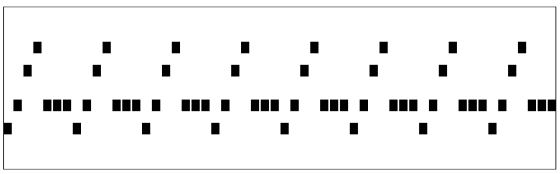
The method developed in this article can also be used to discover patterns that are not fully compact in X. In some cases, this may allow discovering patterns that were



(a) $(P, O_{L=1.75})$ pair where the pattern P is discovered using the erosion by $O_{L=1.75}$



(b) $(P', O'_{L=1.75})$ pair where the pattern P is discovered using the erosion by $O'_{L=1.75}$



(c) $\gamma_{O_{L=1.75}}(X_1) = O_{L=1.75} \oplus P = O'_{L=1.75} \oplus P'$

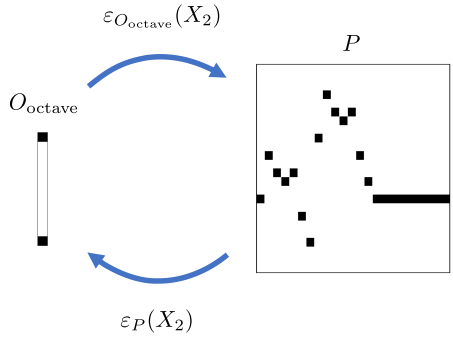
Figure 12. Discovery of repeated patterns by learning the onsets with the exact repetition constraint when $L = 1.75$. Because $|\gamma_{O_{L=1.75}}(X_1)| = |P'| \times |O'_{L=1.75}|$, we conclude that $(P', O'_{L=1.75})$ is a non-redundant pair.

not clearly discernible when listening. For example, let X be the main musical theme of "Piano Phase" by Steve Reich, represented in Figure 14(a). In this case X is non-periodic, in the sense that $\nexists S \subsetneq X$ such that S is not a singleton and $\gamma_S(X) = X$. However, the method based on mathematical morphology proposed in this article allows us to better understand this musical theme.

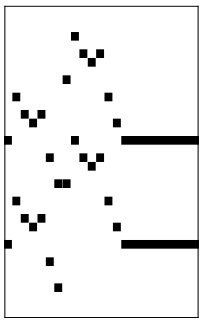
By trying to learn a family of structuring elements that allows X to be describe with morphological operations, we notice that the pairs of musical patterns and onsets (P, O) and (P', O') represented in Figure 14 satisfy the following equality:

$$X = \gamma_P(X) \cup \gamma_{P'}(X) \tag{30}$$

with links between the musical patterns and their onsets, meaning that they are both

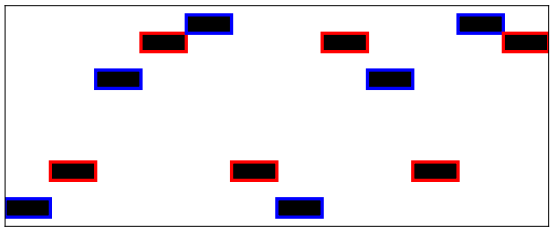


(a) (P, O_{octave}) pair where the pattern P is discovered using the erosion by O_{octave}

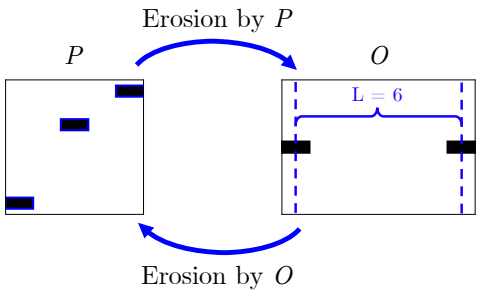


(b) $\gamma_{O_{\text{octave}}}(X_2) = O_{\text{octave}} \oplus P$

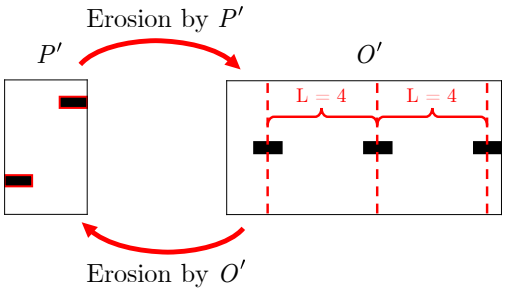
Figure 13. Discovery of repeated patterns by learning the onsets with octave transposition constraint. Because $|\gamma_{O_{\text{octave}}}(X_2)| = |P| \times |O_{\text{octave}}|$, we conclude that (P, O_{octave}) is a non-redundant pair.



(a) X



(b) (P, O) pair with $\varepsilon_P(X) = O$ and $\varepsilon_O(X) = P$



(c) (P', O') pair with $\varepsilon_{P'}(X) = O'$ and $\varepsilon_{O'}(X) = P'$

Figure 14. Musical theme of "Piano Phase" (a) composed of two MTEC conjugate pairs (P, O) and (P', O') of musical patterns and their onsets. The musical theme can be described with morphological operations: $X = (P \oplus O) \cup (P' \oplus O')$.

MTEC conjugate pairs:

$$\varepsilon_P(X) = O \qquad \varepsilon_O(X) = P \qquad \varepsilon_{P'}(X) = O' \qquad \varepsilon_{O'}(X) = P' \tag{31}$$

It is then possible to represent the theme of "Piano Phase" by the following equation:

$$X = (P \oplus O) \cup (P' \oplus O') \tag{32}$$

7. Summary, conclusion, and future work

In this section, we summarize the principal definitions and results introduced in this paper for discovering musical patterns using mathematical morphology. We then conclude this article and propose some directions for future work to extend this approach.

7.1. *Summary*

We have proposed an original approach based on mathematical morphology for discovering musical patterns using a multidimensional representation of music. To do this, we have introduced several definitions which are summarized in Table 1. The fundamental definition provided here is the definition of onsets using morphological erosion, which allows us to distinguish the role of patterns and onsets for the discovery of musical patterns. We then defined a fully compact pattern to be a convex set in musical data, i.e. without holes, which makes the link with the topology present in image processing, where morphological operators are mostly applied. In particular, we proved that with this definition, the problem of discovering a musical pattern from its onsets is satisfied. Finally, we characterized the salient pairs of musical patterns and their onsets as MTEC conjugate and non-redundant, which allows us to discover relevant pairs without having to discover and analyze all the pairs in musical data.

Table 1. Summary of the principal definitions introduced in this article for discovering musical patterns using mathematical morphology, where $P, O, X \subseteq \mathbb{R}^n$.

Definitions	**Formulation using mathematical morphology**
Onsets of P in X	$O = \varepsilon_P(X)$
Discovering pattern from its onsets	$P = \varepsilon_O(X)$
P is *fully compact* in X	$\mathrm{CH}(P) \cap X = P$
MTEC conjugate pair (P, O) in X	$P = \varepsilon_O(X)$ and $O = \varepsilon_P(X)$

The main contribution of this article consists of several mathematical results, which are summarized in Table 2. The majority of these results can be interpreted musically and provide a better understanding of the task of discovering musical patterns.

7.2. *Conclusion and future work*

In this article, we have developed an original application of mathematical morphology to the discovery of musical patterns and demonstrated the relevance of basic morphological operators for this task. We proved several lemmas and theorems that provide a better understanding of the tools used in the musical pattern discovery task. Therefore, the developed method is interesting because it starts from musical problems as a main motivation, i.e. the results and assumptions of the theorems are designed to be applied to musical data, and produces mathematical theoretical results that can also be applied

Table 2. Main results proved in this article that provide an original approach based on algebraic formalism for discovering musical patterns, where $P, O, X \subseteq \mathbb{R}^n$ such that $P \subseteq X$ and $O = \varepsilon_P(X)$.

Name	**Results expressed with mathematical morphology**
Lemma 4.3	$P \subseteq \varepsilon_O(X)$
Theorem 4.5	P is fully compact in $X \Rightarrow P = \varepsilon_O(X)$
Lemma 4.6	$\varepsilon_O(X) = \varphi_O(P)$
Lemma 4.7	$\varepsilon_O(X) \subseteq X$
Theorem 4.11	P is MTEC $\iff P = \varepsilon_O(X)$
Corollary 4.12	P is MTEC $\iff \exists S : P = \varepsilon_S(X)$
Theorem 5.2	$P' = \varepsilon_O(X) \Rightarrow O = \varepsilon_{P'}(X)$
Corollary 5.3	$\exists! P' : (P', O)$ MTEC conjugate
Corollary 5.4	$\nexists Q$ MTEC $: P \subsetneq Q \subsetneq \varepsilon_{\varepsilon_P(X)}(X)$
Lemma 5.6	(P, O) MTEC conjugate $\Rightarrow \gamma_P(X) = \gamma_O(X)$

to other types of discrete set data. For future work, we believe that this method can be generalized to discover repeating patterns and their variations given a certain approximation. To do this, we can use the main idea of Karvonen and Lemström (2008), which is to apply a morphological dilation to musical data to match the patterns with an approximation. For example, add a harmonic tolerance by dilating musical data with $\{(0,0,0),(0,1,0)\}$ as structuring element in order to obtain an approximation of one semi-tone to discover the musical patterns (representing the data in $\mathbb{R}^3$, as explained in Section 6.1, with the pitch as second coordinate). Moreover, it would be appropriate to apply these results to large databases. In addition to the discovery of patterns, this would provide an analysis of the music data, for example by adjusting the method for the segmentation task (the dilation of patterns and their onsets can describe musical sections), and also for the meter detection task (using the distance between the points of the onsets). We also believe that the results developed in this article can be used on audio data by applying a suitable transformation beforehand. Finally, we expect in future research that the use of mathematical morphology to symbolic representations of music will be a tool developed for both musical analysis and generation, while providing interesting and original mathematical results.

Disclosure statement

No potential conflict of interest was reported by the authors.

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